

Biometric Authentication Mechanisms

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Abstract—*Biometric credentials* are prevalent in modern authentication: Sensors evaluate a measurement and output *accept* if it is above a *match threshold* or *reject* otherwise. A large body of work improves the accuracy of biometric sensors, but the state-of-the-art uses a heuristic threshold and does not consider their integration into *authentication mechanisms*. Recent works design authentication mechanisms to maximize the probability that the mechanism accepts a user and rejects an attacker, but treat biometric credentials as *atomic* (perfect match or reject, like passwords).

We model authentication mechanisms with both biometric and atomic credentials and prove that, in general, the optimal threshold differs from the common heuristic. Moreover, the

optimal threshold depends on the integration with other credentials. More surprisingly, combining a biometric credential with an atomic one always improves the success probability over using the atomic credential alone. This holds even if the biometric credential is inaccurate. Using real biometric datasets, we quantify the gain from optimizing the threshold across different mechanisms.

The implication is that biometric sensor vendors should change their interface so integrators can configure thresholds, resulting in better security of deployed authentication mechanisms.

1. Introduction

Biometric credentials are prevalent in modern authentication. However, in the early days of the Internet, biometric credentials were not applicable and therefore, credentials were typically *atomic*, a credential was either presented correctly or not according to a perfect match criterion. Examples include passwords [1], one-time passwords (OTPs) [2] and physical devices [3]. While such credentials are simple to verify, they are vulnerable to various attacks, such as password leaks, account recovery attacks, and device theft. Consequently, biometric credentials are increasingly adopted in modern user devices, with access to most smartphones relying on face or fingerprint verification [4].

Previous work (§2) has conducted extensive analysis of biometric credentials in isolation, primarily focusing on reducing the *false acceptance (FAR)* and *false rejection (FRR) rates* [5], [6], [7]. Sarkar et al. [8] introduce the Equal Error Rate (*EER*, where $FAR = FRR$), a heuristic threshold of biometric sensors. While these works widely cover biometric credentials, none of them analyze their integration into an authentication mechanism. Other works analyze authentication mechanisms as predicates of credential possession [9], [10], [11], [12], where each credential is treated as atomic. But they neglect the potential due to the special properties of biometric credentials.

In this work, we study how to design mechanisms with biometric credentials. We analyze how biometric credentials influence the authentication mechanism's success with and without atomic credentials. First, we present a refined definition of the authentication problem with biometric credentials (§3). The system comprises an authentication mechanism and two players: a *user* and an *attacker*. Both players try to satisfy the mechanism by providing a set of credentials.

Biometric credentials are evaluated by a sensor. The biometric sensors obtains a *sample*, calculates a similarity

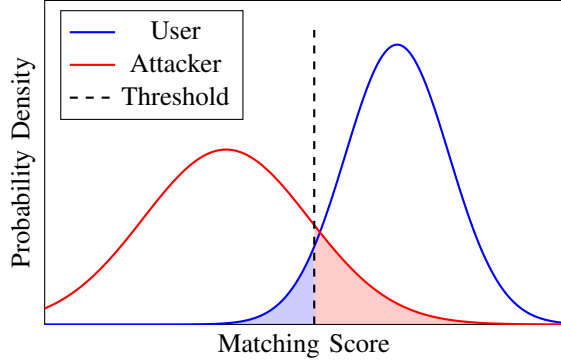


Figure 1: User and Attacker score distributions with *FAR* and *FRR* areas appear in red and blue respectively.

score between that sample and a *reference* stored in advance, and outputs *accept* if the score is above a set *match threshold* or *reject* otherwise. A *genuine distribution* $f_U(s)$ denotes the score distribution of samples that belong to the legitimate user, whereas the *impostor distribution* $f_A(s)$ denotes the score distribution of samples that belong to the attacker, as shown in Figure 1. These distributions establish two error metrics known as *FAR* and *FRR*.

We examine distribution functions that approximate a biometric credential score distribution. We characterize each player by their score distribution. Intuitively, genuine users are expected to produce higher matching scores than attackers. Therefore, the user score distribution is expected to be shifted toward higher matching scores compared to the attacker distribution. Otherwise, the attacker would be more likely to produce scores that lead to successful authentication, rendering the biometric credential ineffective. To define the *state probabilities* of the biometric credential, we calculate the *FAR* and *FRR* metrics which are functions of the match threshold. An optimal mechanism has the maximum probability of success, which depends on the number of credentials and their state probabilities.

We seek to identify optimal mechanisms. Thus, we start by optimizing a single biometric credential mechanism (§4). To define the biometric credential state probabilities, we calculate *FAR* and *FRR* metrics for each biometric credential approximation. Then, we derive closed-form expressions for the mechanism’s success to identify the optimal threshold of the biometric credential. We then symbolically calculate the optimal and *EER* thresholds and show that in general, the optimal threshold differs from the commonly used *EER* threshold. To understand the potential improvement of deviating from the *EER*, we link the likelihood of an attacker producing higher matching scores to the gap between the maximum success and the success at *EER*. Our findings reveal that the more likely an attacker is to produce higher matching scores s , and the more asymmetric the user and attacker distributions become, the greater the benefit of using the optimal threshold instead of *EER*.

Having characterized the optimal threshold of a single biometric credential mechanism, we extend our analysis to

a more complex case that combines two credentials (§5). These may include two biometric credentials, as in airport or border-control systems, or one biometric and one atomic credential, as in cloud service authentication. For example, Google [13] assumes (1) Possession of a trusted device like a phone or computer (Atomic credential) and (2) Authentication via biometric credentials [14], e.g., face or fingerprint.

To evaluate a mechanism comprising two biometric credentials, we consider first and second order approximations and use a numerical optimization algorithm to find the optimal thresholds for each biometric credential. Our results show that the optimal thresholds of the biometric credentials in such a mechanism are not, in general, the same as the optimal thresholds if used individually. This difference arises as we optimize each credential with respect to the entire mechanism, and not only with respect to the credential itself. To demonstrate the potential impact, we present a synthetic experiment of two sampled score distributions and observe a reduction of over 66% in the failure rate, when comparing the *EER* and the optimal threshold.

We then turn to analyze mechanisms that integrate a biometric and an atomic credential. Trying to identify the impact of an atomic credential on the optimal match threshold, we derive closed-form expressions for the mechanism’s success probability. We symbolically calculate the optimal and *EER* thresholds and find that the optimum depends on the values of the atomic credential. These expressions reveal that the mechanism’s success can primarily depend on either an atomic credential or a biometric credential, depending on the atomic credential probability values. For example, in an *AND mechanism*, the mechanism grants access to a user if and only if all credentials are available to the user and at least one credential is unavailable to the attacker. The derived relation for this mechanism shows that when the atomic credential has a high probability of being safe, the mechanism’s success probability depends primarily on the atomic credential. In this case, even biometric samples that closely match the user reference have only a limited impact on the overall mechanism success probability; conversely, for other probability values of an atomic credential, success may depend more strongly on the biometric credential.

Motivated by this observation, we analyze whether the integration of a biometric credential with an atomic credential will always enhance the mechanism’s success probability (§6). First, we link the biometric credential to the improvement it provides to the mechanism’s success probability when it comprises an atomic credential. This link shows that when an atomic credential has zero *loss* or *leak* state probabilities, it is not worthwhile to integrate, as the atomic credential mechanism dominates the mechanism success probability. Then, we characterize the conditions under which integrating a biometric and an atomic credential improves the mechanism’s success probability.

We introduce a necessary and sufficient condition for an *asymptotically separable* biometric credential: for every score above or below a match threshold, the error rates converge to zero. We prove that every *asymptotically separable* biometric credential integrated with an atomic credential

that has non-zero probabilities for the states of *safe* and either *loss* or *leak*, improves the mechanism success probability. However, biometric credentials often exhibit a more common and weaker behavior [15], [16]. Thus, we introduce *asymptotic dominance*: where the genuine distribution dominates the impostor distribution by a factor c for every score above a specific threshold, or the impostor dominates the genuine by a factor c for every score below a specific threshold. We show that this condition is sufficient for a biometric credential to improve the mechanism’s success probability when integrated with an atomic credential.

We quantitatively demonstrate our theoretical analysis using public biometric data and a standard algorithm for fingerprint verification [17] (§7). We use a dataset of fingerprints from multiple users and evaluate matching scores by comparing samples of one user against those of other users. When calculating the matching score produced when different players attempt to authenticate as one another, we observe that the algorithm effectively extracts minutiae and the resulting scores are low. This results in a low *EER* value and a small overlapping area between the user and the attacker distributions. Thus, no significant improvement is observed when comparing the mechanism’s success under the *EER* threshold and the optimal threshold. This indicates that the *EER* threshold may be sufficient and satisfactory when different users try to authenticate as one another.

However, since an attacker does not attempt to authenticate using their own biometric features, the relevant threat model involves attempts to physically impersonate a legitimate user. We conduct a spoofing attack evaluation using the LivDet database [18], [19], [20], [21], which includes both genuine fingerprint scans and corresponding spoofing attempts. These spoof samples produce matching scores that closely resemble those of the legitimate user. Integrating the biometric credential into a mechanism comprising an atomic credential, reduces the mechanism’s failure probability by 30-70% compared to its failure probability under the *EER* threshold. Furthermore, the results show a 50% reduction in the mechanism’s failure probability compared to a mechanism based solely on an atomic credential, demonstrating the benefit of integrating the biometric credential.

In summary (§8), our contributions are both theoretical and practical: (1) Formalization of the mechanism’s success function composed of biometric and atomic credentials; (2) Symbolic analysis of the optimal threshold for biometric and atomic authentication mechanisms. (3) Characterization of the conditions under which biometric credential integration enhances the mechanism’s success; (4) Proof that any biometric credential, under a specified condition, always enhances the mechanism’s success. (5) Empirical analysis using real biometric data. (6) Direct implication that biometric sensor vendors should allow configure thresholds for immediate security improvements.

2. Related Work

Authentication is a constant active and prominent research area, with prior work broadly categorized into

two main aspects: credentials and mechanisms. Credentials are commonly manifested as passwords [1], One-Time-Password (OTP) [2], physical devices [3] or biometric features such as fingerprints, face recognition or iris [16].

Extensive work has evaluated and improved the accuracy of biometric systems [5], [7], [18], [19], [20], [21], [22]. Prabhakar et al. [6] examine the tradeoff between two key error metrics of a biometric system, namely False Acceptance Rate (*FAR*) and False Rejection Rate (*FRR*). Sarkar et al. [8] define the term Equal Error Rate (*EER*), the threshold where *FAR* and *FRR* are equal. They heuristically suggest it as the threshold for biometric credentials. However, these works do not define a success criterion for authentication mechanisms, nor do they explore how the presence of additional credentials can influence the optimal matching threshold of the biometric system. Hong et al. [23] integrate fingerprint and face recognition, each one operates independently, using an *AND mechanism*. They design a decision fusion protocol that makes a decision considering the *FAR* error metric only. However, they do not explore how one biometric credential affects the other for optimal usage, nor do they analyze the integration of biometric credentials along with other credential types within an authentication mechanism.

An authentication mechanism is a function that evaluates a set of credentials provided by a player [12] and output accept or reject [24], [25]. Recent work has explored authentication mechanism defined as predicate of credential possession. Maram et al. [9] study interactive authentication mechanisms, where the user and attacker interact with the authentication mechanism. Mouallem et al. [10] compare mechanisms’ success and aim to identify mechanisms with maximal success when composed of a large number of homogeneous credentials. Eyal [11] analyzes mechanisms modeled as Boolean functions of the credentials. However, these works are partially applicable to our setting. While we adopt their mechanism’s success definition and the way they define the credential possession states, they model biometric credentials as atomic credentials neglecting the tuneable threshold.

3. Model

We formalize the authentication problem by specifying an execution and the system participants (§3.1), credential types and their characteristics (§3.2), and the authentication mechanism and its success (§3.3).

3.1. System Participants

The system comprises an authentication mechanism M and two players: a user U and an attacker A . Each player interacts with the mechanism M by *presenting credentials*. We abstract away the concrete implementation of a credential and model only its presentation to the mechanism. Denote the set of credentials by

$$\mathcal{C} = \{c_1, c_2, \dots, c_n\}$$

The system runs an execution, which describes the interaction between the players and mechanism until the mechanism makes a decision. In an execution, each player P may present a credential c_i to the mechanism by calling a random function $\text{present}_P(i)$. The outcome of the *presentation function* determines whether the credential is available to the player or not, and the mechanism makes a decision based on this outcome. We define the availability of credentials as

Definition 1. (Availability Vector) An availability vector specifies, for each credential, whether a player successfully presented each credential or not. Denote by σ^U , and σ^A the availability vectors of the user and of the attacker, respectively.

For a player $P \in \{U, A\}$, the entry $\sigma_i^P = 1$ means that P successfully presented credential c_i , and $\sigma_i^P = 0$ means that it is not. The outcome of the presentation function determines whether $\sigma_i^P = 1$ or $\sigma_i^P = 0$. Denote the set of credentials available to player P by

$$C_P(\sigma) = \{c_i \in \mathcal{C} \mid \sigma_i^P = 1\}.$$

Each credential can be either: *Safe*: Only available to the user, *Lost*: Not available to either player, *Leaked*: Available to both players, or *Stolen*: Only available to the attacker. A scenario σ is the vector of states of each credential in the mechanism. We denote by σ_i the state of credential c_i in scenario σ . Each state corresponds to the outcome of the presentation functions for a user and an attacker. Specifically, the pair $(\text{present}_U(i), \text{present}_A(i))$ determines whether the credential is *Safe*, *Loss*, *Leak* or *Stolen*. Therefore, the state probabilities are

$$P_{\text{safe}} = \Pr[\text{present}_U(i) = 1 \wedge \text{present}_A(i) = 0].$$

$$P_{\text{loss}} = \Pr[\text{present}_U(i) = 0 \wedge \text{present}_A(i) = 0].$$

$$P_{\text{leak}} = \Pr[\text{present}_U(i) = 1 \wedge \text{present}_A(i) = 1].$$

$$P_{\text{stolen}} = \Pr[\text{present}_U(i) = 0 \wedge \text{present}_A(i) = 1].$$

3.2. Credentials

A credential may correspond to secret data, such as a password or one time password (OTP); a physical object, such as a phone or security key. We model these credentials as atomic: the player either presents the credential correctly or not in a perfect match criterion.

Definition 2. (Atomic Credential) If a credential is atomic, Denote by c_i^* the correct credential stored in advance, then the presentation function $\text{present}_P(i)$ defined by

$$\text{present}_P(i) = \begin{cases} 1 & c_i = c_i^* \\ 0 & \text{otherwise} \end{cases}$$

Alternatively, a credential may correspond to a biometric property, such as a fingerprint or face. The biometric sensor evaluates a measurement of the biometric property and produces a score s . The user distribution $f_U(s)$ denotes the score distribution of samples from the user, whereas the

attacker distribution $f_A(s)$ denotes the score distribution of samples from the attacker.

Definition 3. (Biometric Credential) If a credential is biometric, the presentation function $\text{present}_P(i)$ is defined by $f_U(s)$, $f_A(s)$ and a match threshold t . For a player $P \in \{U, A\}$, the presentation function draw random value for the score s from the distribution $f_P(s)$, and outputs 1 if $s \geq t$ and 0 otherwise.

$$\text{present}_P(i) = \begin{cases} 1 & s \geq t \\ 0 & \text{otherwise} \end{cases}$$

The user and attacker distributions establish two error metrics: *FAR* and *FRR* both are functions of the match threshold t .

$$FAR(t) = \int_t^{a_2} f_A[s]ds \quad (1)$$

$$FRR(t) = \int_{u_1}^t f_U[s]ds \quad (2)$$

We use the *FAR* and *FRR* to find the probability of the biometric credential availability states. The user has access to the biometric credential when the output of the biometric sensor is *accept*, which occurs with probability $1 - FRR(t)$. The attacker has access to the biometric credential when the output of the biometric sensor is *accept*, which occurs with probability *FAR*(t). Thus, the probabilities of the four states are

$$P_{\text{loss}} = FRR(t) \cdot (1 - FAR(t)),$$

$$P_{\text{leak}} = FAR(t) \cdot (1 - FRR(t)),$$

$$P_{\text{stolen}} = FRR(t) \cdot FAR(t), \text{ and}$$

$$P_{\text{safe}} = 1 - P_{\text{loss}} - P_{\text{leak}} - P_{\text{stolen}} = (1 - FAR(t)) \cdot (1 - FRR(t)). \quad (3)$$

3.3. Mechanism

An authentication mechanism M receives the *evaluation* of the presentation function $\text{present}_P(i)$ for each player P and outputs a *decision*.

Definition 4. (Authentication Mechanism) An authentication mechanism M is a predicate that takes as input:

- $C_P(\sigma)$ the set of credentials available to player P in scenario σ (perhaps an empty set, represented by $\emptyset$).

and outputs either 1 meaning that the mechanism accepts the player or 0, meaning that the mechanism rejects the player.

We are now ready to define the mechanism's success. The mechanism is successful in a scenario σ if it grants access to the user only.

Definition 5. (Mechanism Success) A mechanism is successful in a scenario σ if and only if the authentication

mechanism decision $\text{decide}_M(C_P(\sigma))$ equals 1 for the user only, that is

$$\text{decide}_M(C_U(\sigma)) = 1 \wedge \text{decide}_M(C_A(\sigma)) = 0.$$

The mechanism's success probability is thus

$$P_{\text{Success}}(M) = \Pr [\text{decide}_M(C_U(\sigma)) = 1 \wedge \text{decide}_M(C_A(\sigma)) = 0].$$

For example, in an *AND mechanism* [11], the mechanism grants access to a player if and only if all credentials are available to the player. In an *OR mechanism*, the mechanism grants access to a player if at least one credential is available to the player.

4. Mechanisms

We explore optimal mechanism success probability with a single biometric credential. We start with first (§4.1) and second (§4.2) order approximations.

4.1. Uniform Probability Biometric Credentials

We consider uniform distribution as a first-order approximation of the biometric credential distributions. Denote by u_1 and u_2 the user's PDF bounds and by a_1 and a_2 the attacker's PDF bounds as shown in Figure 2. The probability distribution functions for the user and the attacker, f_U and f_A are thus:

$$f_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \leq s \leq u_2 \text{ and}$$

$$f_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \leq s \leq a_2.$$

The error metrics *FAR* and *FRR* are both functions of the match threshold t and given by Equations 1 and 2. Thus:

$$FAR(t) = \int_t^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} 1 & \text{if } t \leq a_1 \\ \frac{a_2 - t}{a_2 - a_1} & \text{if } a_1 < t \leq a_2 \\ 0 & \text{if } t > a_2 \end{cases}$$

and

$$FRR(t) = \int_{u_1}^t \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } t \leq u_1 \\ \frac{t - u_1}{u_2 - u_1} & \text{if } u_1 < t \leq u_2 \\ 1 & \text{if } t > u_2 \end{cases}$$

For a biometric credentials to be useful, users are expected to produce higher matching scores than attackers. Otherwise, the attacker would be more likely to produce scores that lead to successful authentication, rendering the biometric credential ineffective. Thus, we assume that $a_2 < u_2$. There are three distinct cases:

(1) $a_2 < u_1$ (Figure 2a): In this case, the attacker and user distributions are *separated*. For a mechanism M with a single credential c_1 , the mechanism's success probability is simply the probability that the credential is safe:

$$P_{\text{success}}(M^{\text{single}}) = P_{\text{safe}} \quad (4)$$

Any threshold $a_2 \leq t \leq u_1$ yields zero *FAR* and *FRR*, and therefore achieves a mechanism success probability of 1.

(2) $u_1 < a_1 < a_2 < u_2$ (Figure 2b): The attacker distribution bounds lies within the range of the user distribution bounds. The success probability of the biometric credential is given by Equation 3:

$$P_{\text{success}}(t) = (1 - FAR(t)) \cdot (1 - FRR(t))$$

The *FAR* is one for any $t \leq a_1$ and zero for any $t \geq a_2$. Thus:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ (1 - \frac{a_2 - t}{a_2 - a_1}) \cdot (1 - \frac{t - u_1}{u_2 - u_1}), & a_1 < t \leq a_2 \\ 1 - \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ 0, & t > u_2 \end{cases}$$

(3) $a_1 < u_1 < a_2 < u_2$ (Figure 2c): The user and attacker distributions partially overlap.

We maintain the same approach as the case (2) to calculate the probability of success. The *FAR* is one for any $t \leq a_1$ and zero for any $t \geq a_2$, and the *FRR* is zero for any $t \leq u_1$ and one for any $t \geq u_2$. Thus, in this case, we obtain 4 regions of the success probability function:

$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ 1 - \frac{a_2 - t}{a_2 - a_1}, & a_1 < t \leq u_1 \\ (1 - \frac{a_2 - t}{a_2 - a_1}) \cdot (1 - \frac{t - u_1}{u_2 - u_1}), & u_1 < t \leq a_2 \\ 1 - \frac{t - u_1}{u_2 - u_1}, & a_2 < t \leq u_2 \\ 0, & t > u_2. \end{cases}$$

We analyze the probability of success in cases (2) and (3). First, we show that $P_{\text{success}}(t)$ is continuous. Since $P_{\text{success}}(t)$ is continuous on each region of its piecewise definition, it remains only to check the boundary points. At each boundary point, the left-hand and right-hand limits of P_{success} coincide with P_{success} itself; therefore, P_{success} is continuous on its entire domain. Then, we find the optimal threshold that maximizes the mechanism's success probability in each region. The second and third regions of case (2) are equivalent to the third and fourth regions of case (3), thus we analyze them together. In case (3), the success probability function is linearly increasing in $a_1 \leq t \leq u_1$, and therefore its maximum is achieved at the right boundary, $t = u_1$. In both cases, the success probability is linearly decreasing on $a_2 \leq t \leq u_2$, and therefore its maximum is achieved at the left boundary, $t = a_2$. Since $t = u_1$ and $t = a_2$ are the boundary points of the region where *FAR* and *FRR* are strictly between zero and one, the maxima of the adjacent regions are already included in this region. Therefore, it is sufficient to analyze the region where $u_1 \leq t \leq a_2$ in case (3) or $a_1 \leq t \leq a_2$ in case (2). Thus, we find the global maximum by differentiating the success probability function in this region and comparing the resulting critical point with the boundary values.

We differentiate the success function and set it to zero. We find that at: $t = \frac{a_1 + u_2}{2}$ we get $\frac{d}{dt} P_{\text{success}}(t) = 0$ with $\frac{d}{dt} P_{\text{success}}(t) > 0$, For every $t \in (u_1, \frac{a_1 + u_2}{2})$ and $\frac{d}{dt} P_{\text{success}}(t) < 0$, For every $t \in (\frac{a_1 + u_2}{2}, a_2)$. Thus, the

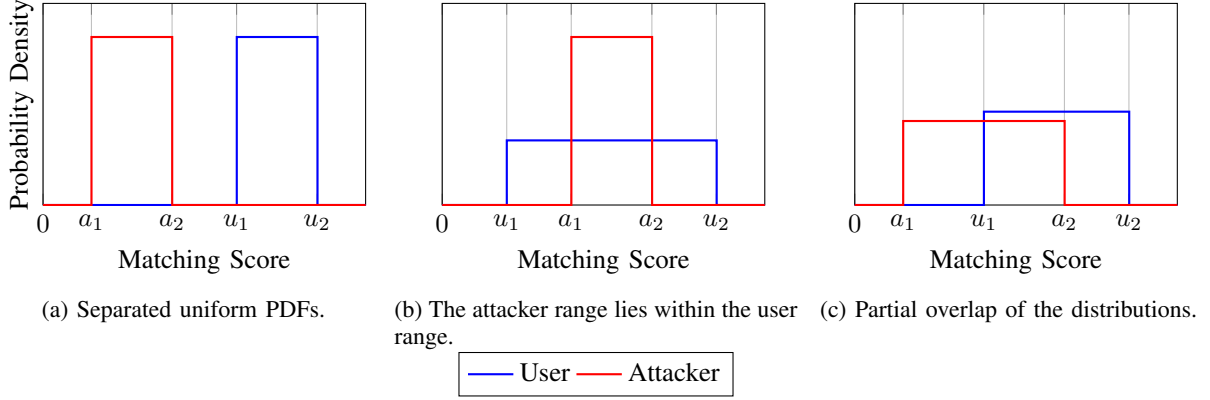


Figure 2: Quantitative illustration of three distinct cases of uniform attacker and user score distributions. In case (1), the distributions are separated. Case (2) illustrate the case where an attacker is fully contained in the user distribution, and (3) illustrate two overlapping configurations.

maximum point is at $t_{\text{Max}} = \frac{a_1 + u_2}{2}$ if: $\frac{a_1 + u_2}{2} \in [u_1, a_2]$ in case (3), or $\frac{a_1 + u_2}{2} \in [a_1, a_2]$ in case (2).

For both cases, the probability of success at the optimal point if the condition is satisfied is

$$P_{\text{success}}(t_{\text{opt}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}.$$

For case (3), if the condition is not satisfied the maximum point is at the boundary of the interval, which is either $t = u_1$ or $t = a_2$. Thus

$$P_{\text{success}}(t_{\text{opt}}) = \max \left\{ \frac{u_1 - a_1}{a_2 - a_1}, \frac{u_2 - a_2}{u_2 - u_1} \right\}.$$

For case (2), if the condition is not satisfied the maximum point is at the boundary of the interval, which is either $t = a_1$ or $t = a_2$. Thus

$$P_{\text{success}}(t_{\text{opt}}) = \max \left\{ 0, \frac{u_2 - a_2}{u_2 - u_1} \right\}.$$

Figure 3 illustrates our analysis of the probability of success for a single biometric credential mechanism as a function of threshold t in the overlapping case. The figure shows that in the second interval the success function is monotonically increasing, while in the fourth interval it is monotonically decreasing. Therefore, the optimal operating point is achieved in the interval where FAR and FRR are strictly between zero and one, which in this scenario is where $u_1 \leq t \leq a_2$.

We calculate the EER point and show that it is not, in general, equal to the optimal point:

$$t_{EER} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2} \neq \left\{ a_1, \frac{a_1 + u_2}{2}, u_1, a_2 \right\} = t_{\text{opt}}.$$

The maximum point represents the optimal threshold of the mechanism. From the optimal success probability expression, we can see that when an attacker distribution spans a wider interval, i.e., when $(a_2 - a_1)$ increases, the mechanism's success probability may significantly decrease.

Figure 4 indicate that when the user and attacker distributions are symmetric, i.e., when the intervals of each

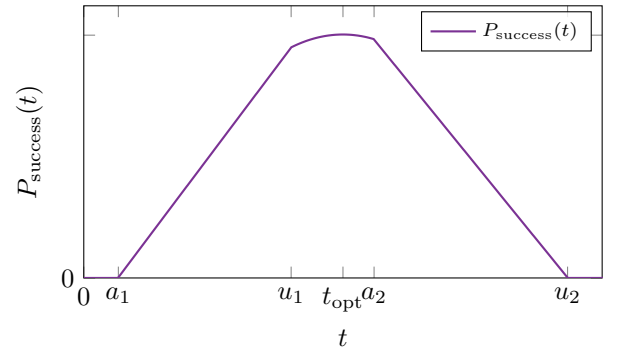


Figure 3: Success probability as a function of the threshold.

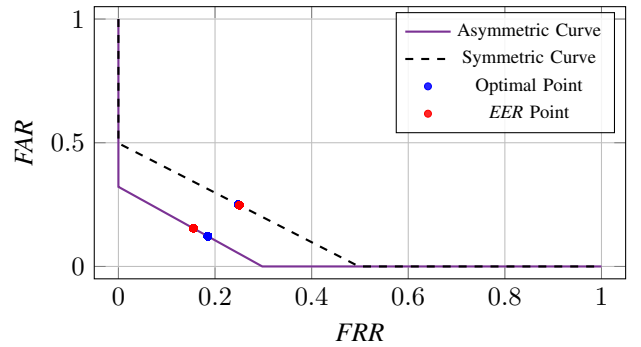


Figure 4: FAR vs. FRR operating curves for the asymmetric and symmetric distributions in (3) case, with optimal and EER points.

distribution are equal: $a_2 - a_1 = u_2 - u_1$, the FAR equals the FRR .

4.2. Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data.

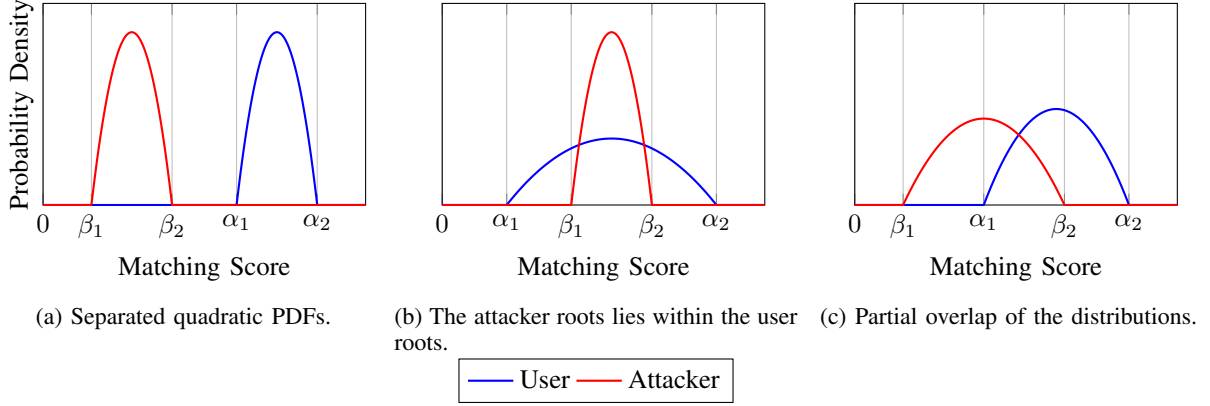


Figure 5: Illustration of three distinct cases of quadratic attacker and user score distributions. In case (1), the distributions are separated. Case (2) illustrates the case where the attacker distribution is fully contained within the user distribution, and case (3) illustrates partial overlap between the two distributions.

To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§7). This choice is also consistent with the findings of Slobodan et al. [15] on Eigenfinger and Eigenpalm feature distributions. Accordingly, we define two probability density functions (PDFs). Let f_U denote the user distribution with roots α_1, α_2 and let f_A denote the attacker distribution with roots β_1, β_2 . Donte by a, b, c and d, e, f the quadratic coefficients of the user and attacker distributions, respectively, hence:

$$f_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \leq s \leq \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$f_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \leq s \leq \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

To reduce the degrees of freedom of the quadratic distributions while maintaining a symbolic analysis, we express the coefficients directly in terms of their roots. The integral of each PDF over its roots interval is set to 1, which enforces the normalization condition. This condition ensures that the resulting error metrics, FAR and FRR are bounded to 1. Under this condition, the coefficients become fully determined. Accordingly, we define the coefficients as follows.

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c) dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f) dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as:

$$f_U(s) = \begin{cases} \frac{6}{(\alpha_1 - \alpha_2)^3} s^2 - \frac{6(\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} s + \frac{6\alpha_1\alpha_2}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \leq s \leq \alpha_2 \\ 0, & \text{otherwise} \end{cases}$$

$$f_A(s) = \begin{cases} \frac{6}{(\beta_1 - \beta_2)^3} s^2 - \frac{6(\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} s + \frac{6\beta_1\beta_2}{(\beta_1 - \beta_2)^3} & \beta_1 \leq s \leq \beta_2 \\ 0, & \text{otherwise} \end{cases}$$

These distributions establish the error metrics, FAR and FRR , as a function of the match threshold t . Denote:

$$A(t) = \frac{2(\beta_2^3 - t^3) - 3(\beta_1 + \beta_2)(\beta_2^2 - t^2) + 6\beta_1\beta_2(\beta_2 - t)}{(\beta_1 - \beta_2)^3},$$

$$U(t) = \frac{2(t^3 - \alpha_1^3) - 3(\alpha_1 + \alpha_2)(t^2 - \alpha_1^2) + 6\alpha_1\alpha_2(t - \alpha_1)}{(\alpha_1 - \alpha_2)^3}.$$

$$FAR(t) = \begin{cases} 1, & t \leq \beta_1, \\ A(t), & \beta_1 < t \leq \beta_2, \\ 0, & t > \beta_2, \end{cases}$$

and

$$FRR(t) = \begin{cases} 0, & t \leq \alpha_1, \\ U(t), & \alpha_1 < t \leq \alpha_2, \\ 1, & t > \alpha_2. \end{cases}$$

Similarly to the analysis of the uniform distribution, the x axis represents the matching score s . Genuine users are expected to produce higher matching scores than attacker, thus we analyze three different cases:

(1) $\beta_2 < \alpha_1$: The two distributions are separated. Any threshold t chosen within the range $\beta_2 \leq t \leq \alpha_1$ yield zero FAR and FRR , and therefore achieves maximal mechanism success as shown in Figure 5a.

(2) $\alpha_1 < \beta_1 < \beta_2 < \alpha_2$: The attacker distribution bounds lies between the range of the user distribution bounds as shown in Figure 5b. The mechanisms probability of success reduces to the safe probability of the biometric credential, which is:

$$P_{\text{safe}} = (1 - \text{FAR}(t)) \cdot (1 - \text{FRR}(t))$$

By setting the $\text{FAR}(t)$ and $\text{FRR}(t)$, we can calculate the probability of success. Denote:

$$G(t) = -\frac{(t - \alpha_2)^2(t - \beta_1)^2}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3} \cdot (2t - 3\alpha_1 + \alpha_2)(2t + \beta_1 - 3\beta_2).$$

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ G(t), & \beta_1 \leq t \leq \beta_2, \\ -\frac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

(3) $\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2$: The user and attacker distributions partially overlap like shown in figure 5c. We maintain same approach as the second case and calculate the probability of success. By setting the $\text{FAR}(t)$ and $\text{FRR}(t)$ into safe probability, the probability of success is:

$$P_{\text{success}}(t) = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ \frac{(t - \beta_1)^2(2t + \beta_1 - 3\beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ G(t), & \alpha_1 \leq t \leq \beta_2, \\ -\frac{(t - \alpha_2)^2(2t - 3\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

The second and third intervals in the second case are identical to the third and fourth intervals in the third case and therefore, we analyze them together. We differentiate $P_{\text{success}}(t)$ and compute its critical points in every interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold t_{opt} .

Before writing the piecewise derivative, we define the polynomial coefficients and the following function:

$$q = 4,$$

$$m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),$$

$$n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,$$

$$\ell = \alpha_1(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)) + \alpha_2^2\beta_2.$$

$$R(t) = -\frac{6(t - \alpha_2)(t - \beta_1)(qt^3 + mt^2 + nt + \ell)}{(\alpha_1 - \alpha_2)^3(\beta_1 - \beta_2)^3}.$$

With these coefficients and function $R(t)$, the derivative of $P_{\text{success}}(t)$ is given by the following piecewise expression:

$$\frac{dP_{\text{success}}}{dt} = \begin{cases} 0, & t < \beta_1 \vee t > \alpha_2, \\ \frac{6(t - \beta_1)(t - \beta_2)}{(\beta_1 - \beta_2)^3}, & \beta_1 \leq t \leq \alpha_1, \\ R(t), & \alpha_1 \leq t \leq \beta_2, \\ -\frac{6(t - \alpha_1)(t - \alpha_2)}{(\alpha_1 - \alpha_2)^3}, & \beta_2 \leq t \leq \alpha_2. \end{cases}$$

In the second interval, the derivative has a single critical point at $t = \beta_1$. Since the derivative is strictly positive for all $t > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $t = \alpha_2$, and the derivative is strictly negative for all $t < \alpha_2$. Therefore, this interval also contains no maximum either.

We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $t = \beta_1$ and $t = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic, reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution can not be simplified further more than a complex expression. Thus, we denote the solution as t_{cubic} .

By examining the sign of the derivative on both sides of the critical point t_{cubic} , we verify that it constitutes a global maximum of $P_{\text{safe}}(t)$. We denote this maximizing point by t_{opt} .

To demonstrate that this optimal threshold differs from the Equal Error Rate (*EER*) threshold, we computed the *EER* point separately as follows:

$$t_{\text{EER}} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2} \neq t_{\text{opt}}$$

We illustrate several user and attacker distributions and present their corresponding probability of success curves, expressed as functions of the threshold t in Figure 6 and Figure 7 respectively. Comparing each case with its corresponding success probability curve, we observe that the optimal threshold t_{opt} becomes more distinct from the *EER* threshold t_{EER} as the variance of the attacker distribution increases. We can also see that the gap between the success probability at the optimal point and the success probability at the *EER* point widens as the attacker's variance increases. This indicates that operating at the optimal threshold can yield a significantly higher success probability compared to operating at the *EER* threshold, especially when the attacker distribution has a higher variance.

Moreover, we illustrate the *FAR-FRR* curves for each pair of user and attacker distributions, and mark both the optimal point and the *EER* point, as shown in Figure 8. We can deduce that as the attacker's variance increases, the optimal point shifts towards α_1 reducing the *FRR* to zero, while the *EER* point remains fixed trying to balance between the *FAR* and *FRR*. This shift indicates that the optimal

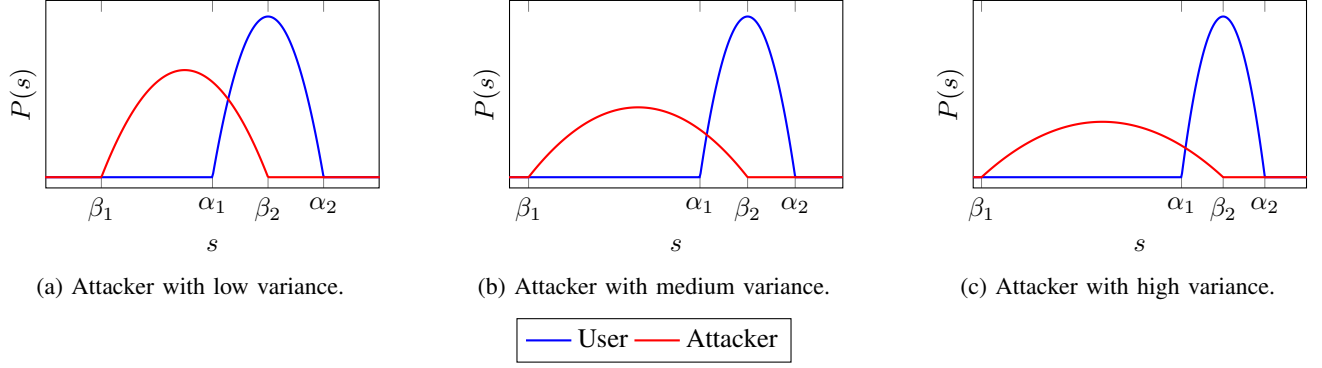


Figure 6: Quadratic user and attacker distributions.

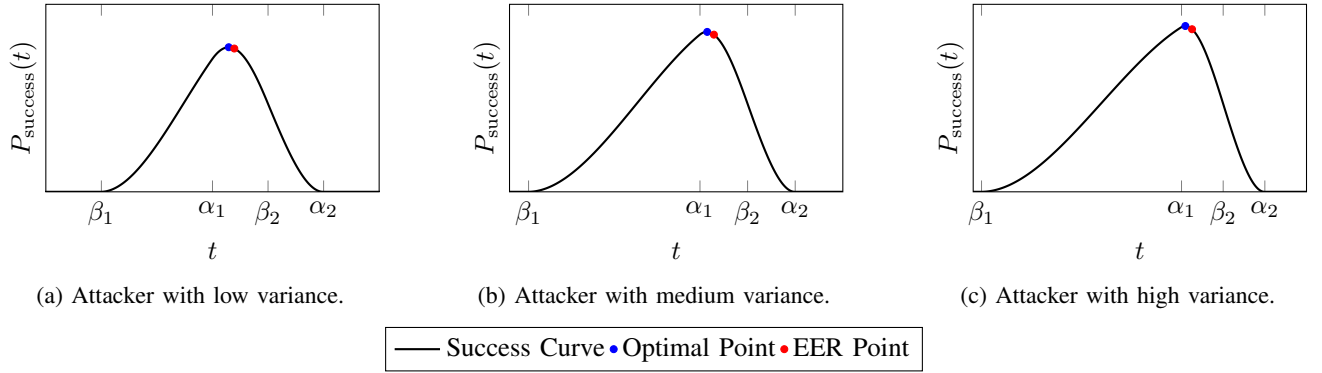


Figure 7: Success probability for three quadratic attacker distributions.

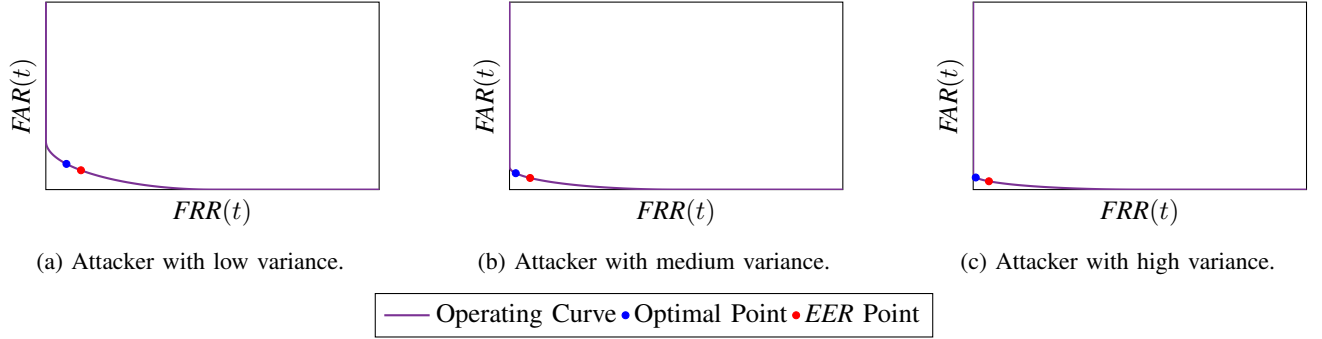


Figure 8: FAR vs. FRR operating curves for different attacker distributions.

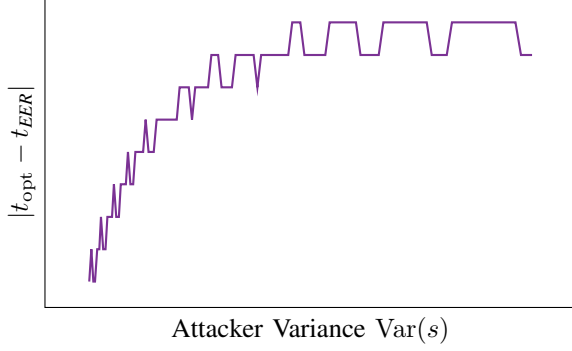
operating point prioritizes minimizing the false rejection rate, rather than balancing both error rates.

To quantify the effect of the variance on the deviation between the optimal operating point t_{opt} and EER point t_{EER} , we numerically computed this gap as a function of the attacker's variance. Decreasing the variance of the attacker achieved by reducing the distance between its roots, corresponds to reduced separability between the two distributions. The simulations reveal that as the variance increases, the gap $|t_{\text{opt}} - t_{EER}|$ grows accordingly, until the operating threshold moves into a regime where it effectively minimizes only one of the error rates, either FAR or FRR , driving the other to zero. This phenomenon is shown in Figure 9a.

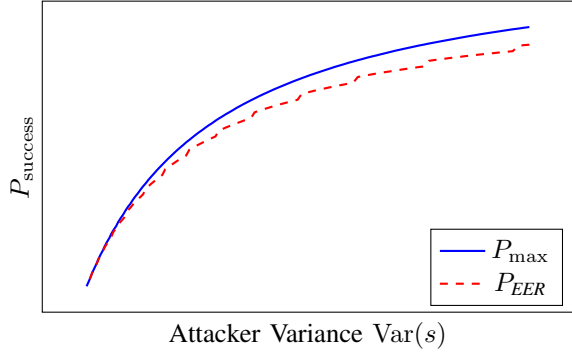
The simulation also indicate that higher attacker variance enlarges the gap between the success probability at the optimal point and the success probability at the EER point as show in Figure 9b. This finding highlights the potential success advantage gained by operating at t_{opt} rather than at the EER point when an attacker has the ability to produce samples with higher scores values.

5. Two Credentials

We explore mechanisms that integrate two independently and identically distributed credentials, derived from either the same or different types of credentials to model real



(a) Threshold gap.



(b) Success at the optimal and *EER* thresholds.

Figure 9: Comparison between the optimal threshold and the *EER* threshold as a function of the attacker's variance.

life implications exist in cloud authentication for example. We numerically analyze two biometric credentials (§5.1) and symbolically analyze one biometric credential and one atomic credential (§5.2).

5.1. 2 Biometric Credentials

To analyze the case of a mechanism M comprised of two biometric credentials, we conduct a numerical study over several probability distribution functions. For each distribution pair, we derive the optimal operating point for both *AND* and *OR* mechanisms by maximizing the respective success probabilities using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the *EER* point, thereby characterizing the conditions under which the *AND* or *OR* mechanisms yields superior performance. We start by considering Gaussian distributions as a third-order approximation of a biometric credential's distribution. We defined the user and attacker distributions as $f_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $f_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We randomly generate the Gaussian parameters to construct a user and an attacker PDF's closely resemble the empirical findings of Slobodan [15]. We focus on the overlapping case between the user and attacker distributions, as this represents the common case among real-world scenarios. For the first credential as $\mu_u = 0.8$, $\sigma_u = 0.12$,

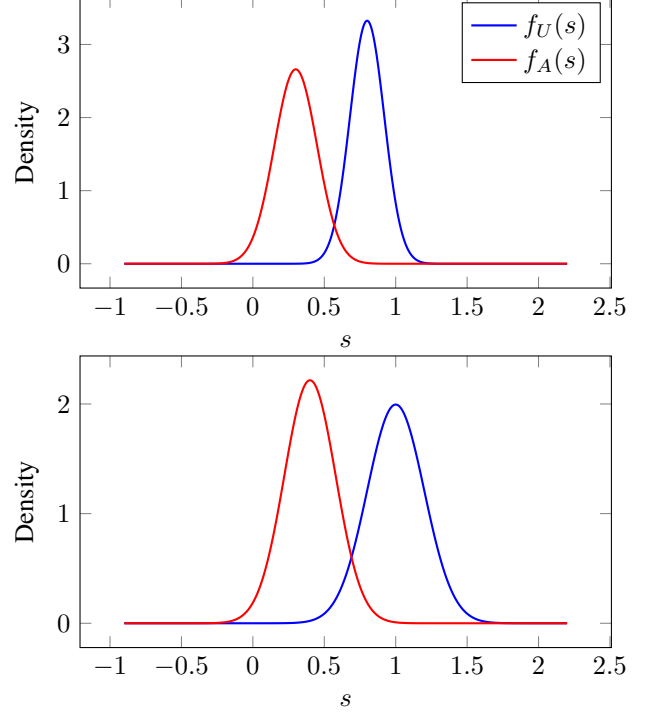


Figure 10: Gaussian user and attacker distributions.

$\mu_a = 0.3$, $\sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4$, $\sigma_a = 0.18$ as shown in Figure 10.

For a mechanism M comprises two independently and identically distributed credentials, derived from either the same or different PDF's, we defined mechanism's success under two scenarios:

a. *AND*: The user has availability to all credentials, while the attacker doesn't have an availability to at least one of the credentials. If both credentials are either in safe state, or one in safe state and the other in leak state the mechanism succeeds. If one of the credentials is either loss or theft, a user U cannot satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{AND}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{leak}}^2 + P_{\text{safe}}^2 \cdot P_{\text{leak}}^1 \quad (5)$$

b. *OR*: The user has availability to at least one credential, while the attacker has no availability to neither of them. If both credential are either in safe state, or one in safe state and the other in loss state the mechanism succeeds. If one of the credentials is either leak or theft, an attacker can satisfy the mechanism and thus the mechanism fails.

$$P_{\text{Success}}(M^{\text{OR}}) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2 + P_{\text{safe}}^1 \cdot P_{\text{loss}}^2 + P_{\text{safe}}^2 \cdot P_{\text{loss}}^1 \quad (6)$$

We calculate the error metrics of each credential c_i which are given by Equations 1 and 2 as a function of threshold t to calculate the state probabilities of each credential as seen in Section 3.2.

$$FAR(t) = \int_{-\infty}^t \frac{1}{\sigma_{a,i} \sqrt{2\pi}} \exp\left(-\frac{(s - \mu_{a,i})^2}{2\sigma_{a,i}^2}\right) ds, \text{ and}$$

$$FRR(t) = \int_t^\infty \frac{1}{\sigma_{u,i}\sqrt{2\pi}} \exp\left(-\frac{(s - \mu_{u,i})^2}{2\sigma_{u,i}^2}\right) ds.$$

We analyze both *AND* and *OR* mechanisms success given by Equations 5 and 6 and find their optimal threshold t , using LBFGS optimization algorithm. We also compute the *EER* point for each credential and find the corresponding success probability at this point.

The optimal operating point for both mechanisms' types differ from the *EER*, yielding higher success probabilities than those obtained at the *EER*. Moreover, we find that the optimal points of both the *AND* and *OR* mechanisms are not necessarily the same as the optimal threshold if used individually. This finding suggest that integrators of a biometric credential should determine its threshold as function of the other credentials integrated into the mechanism. Figure 11 illustrates the *FAR* vs. *FRR* operating curves for both credentials, and marks the optimal points of both *AND* and *OR* mechanisms, as well as the *EER* and the standalone optimum of the biometric credential. The figure highlights the gap between different operating points. In the first credential, the threshold leads to high *FRR* and low *FAR* values, while in the second credential the threshold leads to low *FRR* and high *FAR* values. This demonstrates that the *AND* and *OR* mechanisms balance the two error rates differently across credentials.

We also find that the integration of a biometric credential to a mechanism already comprises a biometric credential, improves both *AND* and *OR* mechanism success. Hence, integrating two biometric credentials into a mechanism improves its success over using them individually. Figure 12 illustrate the success probability of both *AND* and *OR* mechanisms. We fix a credential to its optimum and show how the success probability varies as a function of the thresholds of the other credential.

We numerically analyze first and second order approximations of the biometric credential's distribution and calculate the success probabilities of both *AND* and *OR* mechanisms at different thresholds. We quantify the improvement achieved by integrating two biometric credentials and the improvement of using the optimal threshold over the *EER* threshold. In conclusion, all numerical results are summarized in Table 1.

From Table 1, we observe that the optimal operating point yields a higher success probability than the *EER* point by between 5% to 7%.

5.2. One Atomic and One Biometric Credential

We start by analyzing the case of an *AND* mechanism comprises an atomic credential and a biometric credential. We define the atomic credential with a constant value of each probabilities' states as: P_{safe}^1 , P_{loss}^1 , P_{leak}^1 , P_{theft}^1 corresponding respectively to each one of the states: safe, loss, leak and theft. We define the biometric credential by its PDFs for the user and the attacker, f_U and f_A respectively. We consider uniform distribution as a

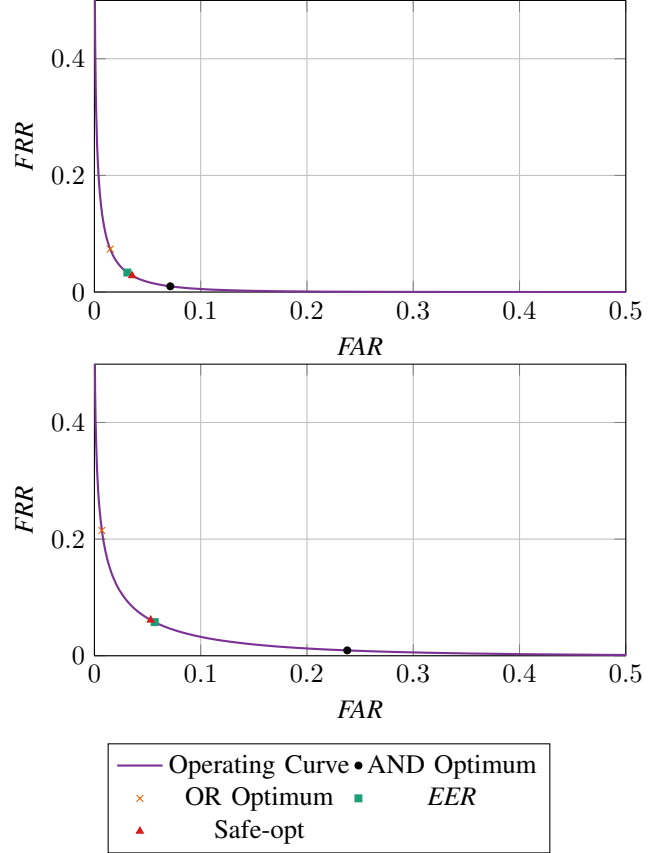


Figure 11: *FAR* vs. *FRR* operating curves for two biometric credentials.

first-order approximation of a biometric credential's distribution. Denote by u_1, u_2 the user's PDF bounds and a_1, a_2 the attacker's PDF bounds, as shown in Figure 2. The state probabilities of the biometric credential are defined by the error metrics *FAR* and *FRR* as done in Section 3.2. Moreover, we consider the overlapping case between the user and attacker distributions, as this represents the common case among real-world scenarios. Thus, we assume that $a_1 < u_1 < a_2 < u_2$. We set the state probabilities of an atomic and a biometric credential into Equation 5. Thus, the probability of success is a function of the threshold t and divided into 5 intervals. Denote:

$$L(t) = \frac{u_2 - t}{u_2 - u_1} \left[P_{safe}^1 + P_{leak}^1 \left(1 - \frac{a_2 - t}{a_2 - a_1} \right) \right].$$

$$P_{success}(t) = \begin{cases} P_{safe}^1, & t \leq a_1 \\ P_{safe}^1 + P_{leak}^1 \left(1 - \frac{a_2 - t}{a_2 - a_1} \right), & a_1 < t \leq u_1 \\ L(t), & u_1 < t \leq a_2 \\ \frac{u_2 - t}{u_2 - u_1} \cdot (P_{safe}^1 + P_{leak}^1), & a_2 < t \leq u_2 \\ 0, & t > u_2. \end{cases}$$

We analyze the success probability $P_{success}(t)$ by partitioning the threshold domain into five intervals.

Distribution	$P_{c_1}(\text{safe})$	$P_{c_2}(\text{safe})$	$P_{\text{AND}}(t_{\text{EER}})$	$P_{\text{OR}}(t_{\text{EER}})$	$P_{\text{AND}}(t_1^{\text{safe}})$	$P_{\text{OR}}(t_2^{\text{safe}})$	$P_{\text{AND}}(t_{\text{opt}})$	$P_{\text{OR}}(t_{\text{opt}})$	Improvement
Gaussian	0.937	0.889	0.911	0.911	0.912	0.911	0.964	0.963	5.90% / 5.66%
Uniform	1.000	0.916	0.956	0.956	0.917	0.999	1.000	1.000	4.63% / 4.61%
Parabolic	0.959	0.887	0.921	0.921	0.920	0.923	0.987	0.985	7.25% / 6.99%

TABLE 1: Mechanism success under two biometric credentials with Gaussian, Uniform and Parabolic distributions at four reference operating points: safe success of a single biometric credential mechanism, EER -pair $(t_1^{\text{EER}}, t_2^{\text{EER}})$, individual optimum pair $(t_1^{\text{safe}}, t_2^{\text{safe}})$, and joint optima (AND at $(t_1^*, t_2^*)_{\text{AND}}$ and OR at $(t_1^*, t_2^*)_{\text{OR}}$). The improvement is measured between the optimal operating point and the EER point.

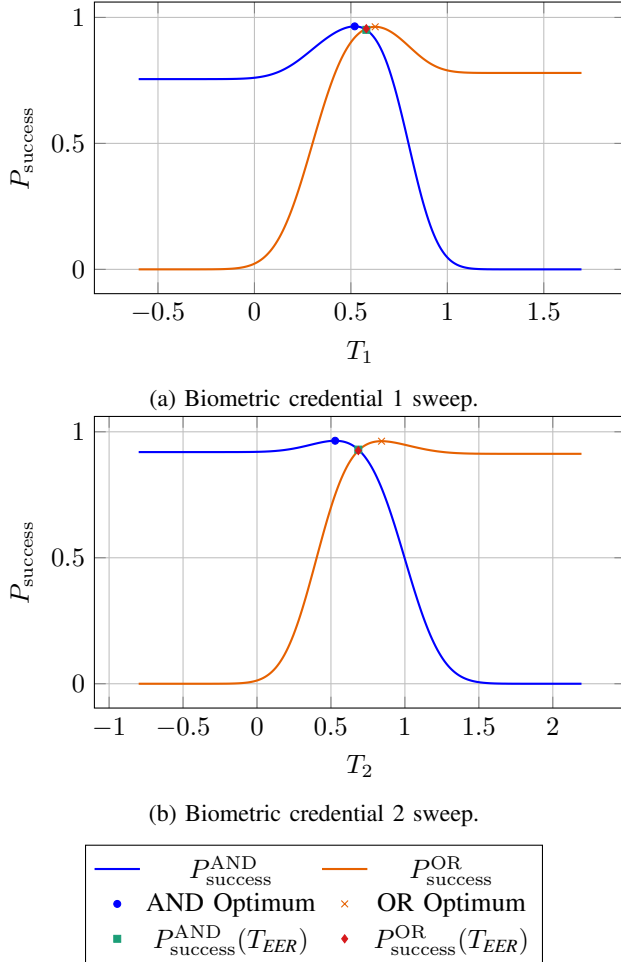


Figure 12: Success probability as a function of the threshold for two biometric credentials.

The first interval, $t \leq a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the FAR is equal to 1, while the FRR is equal to 0. Consequently, the safe probability of the biometric credential is zero, and the success probability converges to P_{safe}^1 . Since P_{safe}^1 is constant in this interval, no interior maximum exists.

The second interval, $a_1 < t \leq u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the user distribution. In this region, the FAR decreases linearly

as a function of t , while the FRR remains equal to 0. As a result, P_{safe}^2 and P_{leak}^2 of the biometric credential are $1 - FAR(t)$ and $FAR(t)$, respectively. In this interval, $P_{\text{success}}(t)$ is a strictly increasing function, as its derivative is a constant positive value, and therefore the maximum is attained at the boundary point: $t_{\text{opt},2} = u_1$.

The third interval, $u_1 < t \leq a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both FAR and FRR vary linearly with t , resulting in a success probability that is quadratic as a function of t . Differentiating $P_{\text{success}}(t)$ yields a single critical point inside the interval,

$$t'_{\text{opt}} = \frac{u_2 + a_2 - (a_2 - a_1) \frac{P_{\text{safe}}^1 + P_{\text{leak}}^1}{P_{\text{leak}}^1}}{2}$$

Since this expression is valid only for the interval $u_1 < t \leq a_2$, the effective maximizer within this interval is obtained by clipping the optimum to the interval boundaries, as follows:

$$t_{\text{opt},3} = \min(a_2, \max(u_1, t'_{\text{opt}})).$$

In particular, if $t_{\text{opt},3} < u_1$, the maximum over this interval is attained at $t = u_1$, whereas if $t_{\text{opt},3} > a_2$, the maximum is attained at $t = a_2$. Evaluating the success probability at the user threshold $t = u_1$ yields

$$P_{\text{success}}(u_1) = P_{\text{safe}}^1 + P_{\text{leak}}^1 \left(1 - \frac{a_2 - u_1}{a_2 - a_1}\right).$$

Since we are analyzing the overlapping case where $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P_{\text{safe}}^1,$$

Demonstrating that the integration of the biometric credential strictly improves the success of the mechanism compared to using the atomic credential alone.

By examining the optimal point in this interval under AND mechanism, we can deduce that as the variance of the attacker distribution increases, the optimal threshold $t_{\text{opt},3}$ shifts toward the lower boundry of the interval, u_1 . Moreover, for an atomic credential with high safe values P_{safe}^1 , and low leak values P_{leak}^1 , the optimal threshold shifts toward the lower boundary as well. Thus, in such case, the success of the mechanisms becomes mainly dependent on the atomic credential. Conversely, an atomic credential with high leak values P_{leak}^1 and low safe values P_{safe}^1 causes the optimal threshold to shift toward the upper boundary of the

interval, a_2 . In this case, the success of the mechanism relies more heavily on the biometric credential.

The fourth interval, $a_2 < t \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the FAR is equal to 0, while the FRR increases linearly with t . Consequently, $P_{\text{success}}(t)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$t_{\text{opt},4} = a_2.$$

Finally, the fifth interval, $t > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $t_{\text{opt}} = t_{\text{opt},3}$ into $P_{\text{success}}(t)$ yields

$$P_{\text{success}}(t_{\text{opt},3}) = \frac{(-a_1(P_{\text{safe}}^1 + P_{\text{leak}}^1) + a_2P_{\text{safe}}^1 + P_{\text{leak}}^1u_2)^2}{4P_{\text{leak}}^1(a_2 - a_1)(u_2 - u_1)}.$$

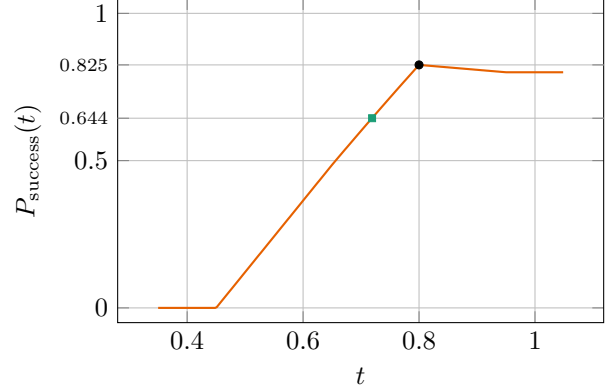
We do similar analysis for the *OR* mechanism. We find the probability of success for the *OR* mechanism. Denote:

$$Q(t) = \frac{t - a_1}{a_2 - a_1} \left[P_{\text{safe}}^1 + P_{\text{loss}}^1 \left(1 - \frac{t - u_1}{u_2 - u_1} \right) \right].$$

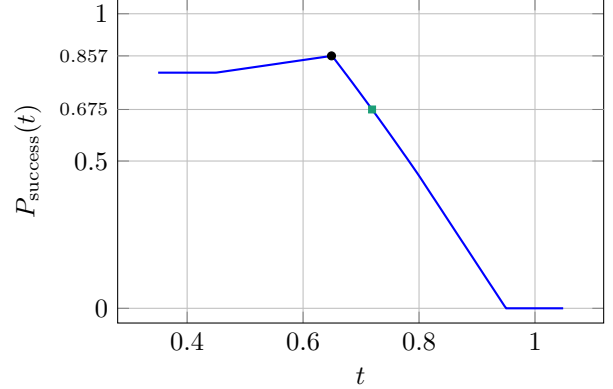
$$P_{\text{success}}(t) = \begin{cases} 0, & t \leq a_1 \\ \frac{t - a_1}{a_2 - a_1} (P_{\text{safe}}^1 + P_{\text{loss}}^1), & a_1 < t \leq u_1 \\ Q(t), & u_1 < t \leq a_2 \\ P_{\text{safe}}^1 + P_{\text{loss}}^1 \left(1 - \frac{t - u_1}{u_2 - u_1} \right), & a_2 < t \leq u_2 \\ P_{\text{safe}}^1, & t > u_2. \end{cases}$$

We illustrate the mechanism's success probability for both *OR* mechanism in figure 13a and *AND* mechanism in Figure 13b. The figure marks the optimal operating point and the *EER* point on each curve demonstrating significant improvement from *EER* using optimal threshold. Moreover, the figure shows that when integrating a biometric credential and determining the threshold t to *EER*, the integration reduces mechanism's success compared to using the atomic credential alone. However, by optimizing the threshold, the integration of the biometric credential improves the mechanism's success over using the atomic credential alone. By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where $u_1 < t \leq a_2$:

$$t_{\text{opt}} = \frac{a_1 \cdot P_{\text{loss}}^1 + u_2 \cdot (P_{\text{safe}}^1 + P_{\text{loss}}^1) - P_{\text{safe}}^1 \cdot u_1}{2P_{\text{loss}}^1}$$



(a) OR mechanism.



(b) AND mechanism.



Figure 13: Success probability as a function of the threshold for the OR and AND mechanisms composed of an atomic credential with state probabilities $P_{\text{safe}}^1 = 0.8$, $P_{\text{loss}}^1 = 0.05$, $P_{\text{leak}}^1 = 0.1$, $P_{\text{theft}}^1 = 0.05$, and a biometric credential.

And the maximum probability of success at this point is:

$$P_{\text{success}}(t_{\text{opt}}) = \frac{(a_1P_{\text{loss}}^1 - u_2(P_{\text{safe}}^1 + P_{\text{loss}}^1) + P_{\text{safe}}^1u_1)^2}{4P_{\text{loss}}^1(a_2 - a_1)(u_2 - u_1)}$$

we calculate the *EER* point and show that the optimal operating point t_{opt} differs from the equal error rate (*EER*) point t_{EER} .

$$t_{\text{EER}} = \frac{a_1u_1 - a_2u_2}{a_1 - a_2 + u_1 - u_2} \neq t_{\text{opt}}.$$

6. Characterizing Success Improvement

We will now characterize the conditions under which integrating a biometric credential into a single atomic credential mechanism, enhances its success. We show that for any mechanism type whose atomic credential characterized by non-zero probabilities P_{safe}^1 and P_{stolen}^1 only, integrating

a biometric credential does not improve mechanism's success (§6.1). Next, we present a necessary and sufficient condition for such improvement (§6.2) and derive a more common criterion that provides a sufficient condition (§6.3).

6.1. No Improvement In Mechanisms Success

Proposition 1. *For any mechanism M comprises an atomic credential with non-zero probabilities P_{safe}^1 and P_{stolen}^1 only, integrating the mechanism with a biometric credential does not enhance its mechanism's success.*

Proof. Integrating a biometric credential into a mechanism M comprises an atomic credential with non-zero probabilities P_{safe}^1 and P_{stolen}^1 only, leads to mechanism success probability described by Equation 5 or Equation 6. Since the state probabilities of an atomic credential are $P_{\text{leak}}^1 = P_{\text{loss}}^1 = 0$, the success probability in both mechanism cases reduces to:

$$P_{\text{success}}(t) = P_{\text{safe}}^1 \cdot P_{\text{safe}}^2(t).$$

Where $P_{\text{safe}}^2(t)$ is the safe probability of the biometric credential as a function of the operating threshold t . However, $P_{\text{safe}}^2(t) \leq 1$ for every value of t . Then $P_{\text{success}}(t) \leq P_{\text{safe}}^1$ and thus, integrating a biometric credential does not enhance the success of the mechanism. $\square$

6.2. Sufficient And Necessary Condition for Integrating Biometric Credentials

First, we define the condition of *asymptotic separability* of a biometric credential. Then, we will show that its a necessary and sufficient condition for enhancing the mechanism's success when integrating a biometric credential.

Definition 6. *A biometric credential with user distribution function $f_U(s)$ and attacker distribution function $f_A(s)$ is said to be asymptotically separable if*

$$\inf_t \min \left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0,$$

where $FAR(t) = \int_t^\infty f_A(s)ds$ and $FRR(t) = \int_{-\infty}^t f_U(s)ds$.

Proposition 2. *For any mechanism M comprises an atomic credential with non-zero probabilities P_{safe}^1 and P_{leak}^1 or P_{loss}^1 , integrating the mechanism with a biometric credential whose user and attacker distributions are asymptotically separable, enhances the success of the mechanism.*

Proof. The success probability of a mechanism comprises an atomic credential only is given by Equation 4. A biometric credential is characterized by four state probabilities which are function of t : $P_{\text{safe}}^2(t)$, $P_{\text{loss}}^2(t)$, $P_{\text{leak}}^2(t)$, $P_{\text{theft}}^2(t)$.

We can describe each one of these probabilities as a function of FAR and FRR as:

$$\begin{aligned} P_{\text{safe}}^2(t) &= 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t), \\ P_{\text{loss}}^2(t) &= FRR(t)(1 - FAR(t)), \\ P_{\text{leak}}^2(t) &= FAR(t)(1 - FRR(t)), \\ P_{\text{theft}}^2(t) &= FAR(t) \cdot FRR(t). \end{aligned}$$

The success probability of the integrated mechanism comprises two credentials is described by Equation 5 and Equation 6. Thus, by placing the biometric credential in each configuration we can express the success probability of the integrated mechanism as a function of the operating threshold t :

$$P_{\text{success}}^{\text{AND}}(t) = (1 - FRR(t)) \left(P_{\text{safe}}^1 + P_{\text{leak}}^1(1 - FAR(t)) \right). \quad (7)$$

$$P_{\text{success}}^{\text{OR}}(t) = (1 - FAR(t)) \left(P_{\text{safe}}^1 + P_{\text{loss}}^1(1 - FRR(t)) \right). \quad (8)$$

To prove the integration of a biometric credential improved mechanism success, we need to prove that there exists an operating threshold t such that either $P_{\text{success}}^{\text{AND}}(t) > P_{\text{safe}}^1$ or $P_{\text{success}}^{\text{OR}}(t) > P_{\text{safe}}^1$. By algebraic manipulation of Equation 7 we will demand that $P_{\text{success}}^{\text{AND}}(t) > P_{\text{safe}}^1$ if and only if:

$$1 - FRR(t) > \frac{P_{\text{safe}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1(1 - FAR(t))} \iff$$

$$FRR(t) < 1 - \frac{P_{\text{safe}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1(1 - FAR(t))} \iff$$

$$FRR(t) < \frac{P_{\text{leak}}^1 \cdot (1 - FAR(t))}{P_{\text{safe}}^1 + P_{\text{leak}}^1(1 - FAR(t))} \iff$$

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1(1 - FAR(t))}$$

Since $P_{\text{leak}}^1 \cdot (1 - FAR(t)) \leq P_{\text{leak}}^1$, a sufficient condition for the above inequality is:

$$\frac{FRR(t)}{1 - FAR(t)} < \frac{P_{\text{leak}}^1}{P_{\text{safe}}^1 + P_{\text{leak}}^1} \quad (9)$$

Similarly, we can deduce that $P_{\text{success}}^{\text{OR}}(t) > P_{\text{safe}}^1$ if and only if:

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1(1 - FRR(t))}$$

Since $P_{\text{loss}}^1 \cdot (1 - FRR(t)) \leq P_{\text{loss}}^1$, a sufficient condition for the above inequality is:

$$\frac{FAR(t)}{1 - FRR(t)} < \frac{P_{\text{loss}}^1}{P_{\text{safe}}^1 + P_{\text{loss}}^1} \quad (10)$$

By the definition of asymptotic separability, there exists an operating threshold t such that:

$$\inf_t \min \left\{ \frac{FAR(t)}{1 - FRR(t)}, \frac{FRR(t)}{1 - FAR(t)} \right\} = 0.$$

Equivalently, $\forall \varepsilon > 0, \exists t$ such that either $\frac{FAR(t)}{1-FRR(t)} < \varepsilon$ or $\frac{FRR(t)}{1-FAR(t)} < \varepsilon$. Thus, we can choose ε that equals the right hand side of either Equation 9 or Equation 10 and by definition we know that a t satisfies the inequality exists. More precisely, if

$$\inf_t \frac{FAR(t)}{1-FRR(t)} = 0,$$

then the OR mechanism composition is preferable, whereas if

$$\inf_t \frac{FRR(t)}{1-FAR(t)} = 0,$$

then the AND mechanism composition is preferable. Therefore, integrating an asymptotically separable biometric credential enhances the success of the mechanism. $\square$

Proposition 3. Suppose there $\exists t$ such that the integrated mechanism M comprises a biometric credential and an atomic credential with non zero probabilities P_{safe} and P_{leak} or P_{loss} achieves a strictly higher success probability than an atomic credential mechanism, i.e.,

$$P_{success}^{integrated}(t) > P_{success}^{atomic}.$$

Then necessarily, the biometric credential is asymptotically separable.

Proof. We prove by contradiction. Assume for the sake of contradiction, that the biometric credential is not asymptotically separable. Then $\exists \varepsilon > 0$ such that $\forall t$ both $\frac{FAR(t)}{1-FRR(t)} \geq \varepsilon$ and $\frac{FRR(t)}{1-FAR(t)} \geq \varepsilon$. Since it's given that the integrated mechanism M achieves a strictly higher success probability than an atomic credential mechanism, then there exists a threshold t such that either: $P_{success}^{AND}(t) > P_{safe}^1$ or $P_{success}^{OR}(t) > P_{safe}^1$. Thus, either Equation 9 or Equation 10 holds for every value of $P_{safe}^1, P_{leak}^1, P_{loss}^1$ in the range of $(0, 1)$. In particular, we can choose $P_{safe}^1, P_{leak}^1, P_{loss}^1$ such that either

$$\frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1} < \varepsilon \text{ or } \frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1} < \varepsilon.$$

Since the equation holds $\forall \varepsilon > 0$, but both $\frac{FAR(t)}{1-FRR(t)} \geq \varepsilon$ and $\frac{FRR(t)}{1-FAR(t)} \geq \varepsilon$ derive by the sake of contradiction assumption, we can deduce that

$$\frac{FRR(t)}{1-FAR(t)} \geq \frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1} \text{ and } \frac{FAR(t)}{1-FRR(t)} \geq \frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1}.$$

Which contradicts both Equation 9 and Equation 10 under the strictly higher success probability assumption. Therefore, the biometric credential must be asymptotically separable. $\square$

We show that *asymptotic separability* of the user and attacker distributions is a sufficient and necessary condition for enhancing the success of a mechanism comprises an atomic credential with non zero probabilities P_{safe}, P_{leak} or P_{loss} by integrating it with a biometric credential.

6.3. A Sufficient Condition For biometric Credentials

We define a *asymptotic dominancy*, which is a more common condition of a biometric credential, and prove this is a sufficient condition for enhancing the success of an atomic credential mechanism.

Definition 7. A biometric credential with user a distribution function $f_U(s)$ and an attacker distribution function $f_A(s)$ is said to be asymptotically dominated if:

$$\forall c > 0 \exists t \text{ s.t. : } \forall s \geq t : f_A(s) \leq c f_U(s)$$

or

$$\forall c > 0 \exists t \text{ s.t. : } \forall s \leq t : f_U(s) \leq c f_A(s)$$

Proposition 4. For any mechanism M comprises an atomic credential and mechanism success probability $P_{success}^{atomic}$, integrating the mechanism with an asymptotically dominated biometric credential improves its achievable success. That is

$$P_{success}^{integrated}(t) > P_{success}^{atomic}.$$

Proof. The success probability of a mechanism comprises an atomic credential only is given by Equation 4. A biometric credential is characterized by four state probabilities which are function of t : $P_{safe}^2(t), P_{loss}^2(t), P_{leak}^2(t), P_{theft}^2(t)$. We can describe each one of these probabilities as a function of FAR and FRR as:

$$\begin{aligned} P_{safe}^2(t) &= 1 - FAR(t) - FRR(t) + FAR(t) \cdot FRR(t), \\ P_{loss}^2(t) &= FRR(t)(1 - FAR(t)), \\ P_{leak}^2(t) &= FAR(t)(1 - FRR(t)), \\ P_{theft}^2(t) &= FAR(t) \cdot FRR(t). \end{aligned}$$

The success probability of the integrated mechanism comprises two credentials is described by Equation 5 and Equation 6. Thus, we can express the success probability of the integrated mechanism as a function of the operating threshold t as shown in Equation 7 and Equation 8. We previously observed that a sufficient condition for improvement of mechanism's success is described by Equation 9 and Equation 10. Thus, we need to show that there exists an operating threshold t such that either:

$$\frac{FRR(t)}{1-FAR(t)} < \frac{P_{leak}^1}{P_{safe}^1 + P_{leak}^1} \text{ or } \frac{FAR(t)}{1-FRR(t)} < \frac{P_{loss}^1}{P_{safe}^1 + P_{loss}^1}$$

We expand left hand side of Equation 9.

$$\frac{FRR(t)}{1 - FAR(t)} = \frac{\int_{-\infty}^t f_U(s)ds}{\int_{-\infty}^t f_A(s)ds}$$

By Definition 7, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \leq t$, $f_U(s) \leq c \cdot f_A(s)$. Therefore:

$$\frac{\int_{-\infty}^t f_U(s)ds}{\int_{-\infty}^t f_A(s)ds} \leq \frac{\int_{-\infty}^t c \cdot f_A(s)ds}{\int_{-\infty}^t f_A(s)ds} = c$$

This is true $\forall c > 0$, thus we can choose c such that satisfies Equation 9. We expand left hand side of Equation 10.

$$\frac{FAR(t)}{1 - FRR(t)} = \frac{\int_t^{\infty} f_A(s)ds}{\int_t^{\infty} f_U(s)ds}$$

By Definition 7, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold t such that for every $s \geq t$, $f_A(s) \leq c \cdot f_U(s)$. Therefore:

$$\frac{\int_t^{\infty} f_A(s)ds}{\int_t^{\infty} f_U(s)ds} \leq \frac{\int_t^{\infty} c \cdot f_U(s)ds}{\int_t^{\infty} f_U(s)ds} = c, \forall c > 0$$

We showed that $\forall c > 0$ either $\frac{FAR(t)}{1 - FRR(t)} < c$ or $\frac{FRR(t)}{1 - FAR(t)} < c$. Therefore, integrating an asymptotically dominated biometric credential with an atomic credential improves mechanism success probability. $\square$

This proof is not trivial, as the integration of another atomic credential to an atomic credential mechanism, does not always enhance its success. Moreover, we have shown when integrating biometric credential into an atomic credential mechanism without adjusting the match threshold t might reduce the mechanism's success probability (§5.2). Thus, Proposition 4 shows that by using an *asymptotically dominated* biometric credential, we can always improve the success of all mechanism's composed of an atomic credential. This condition supports practical scenarios, as we expect that for very low values of score s the attacker distribution dominates the user, vice versa for very high values of s . We supports this claim with real public biometric data in Section 7.

7. Measurements

We now present experiments' results based on fingerprints taken from *FVC2002* database [26] and *FVC2004* database [27]. We use a standard fingerprint verification algorithm *SourceAFIS* created by Vazan [17] for scoring fingerprints out of their images based on minutiae triplets. We use fingerprints, as it is a widely common and used example of a biometric credential in authentication mechanisms. For a test set, we took five templates out of *FVC2002 database1* of user 103 and compare them against the whole database of both *FVC2002* and *FVC2004*. A genuine will be considered the templates of user 103 against its other

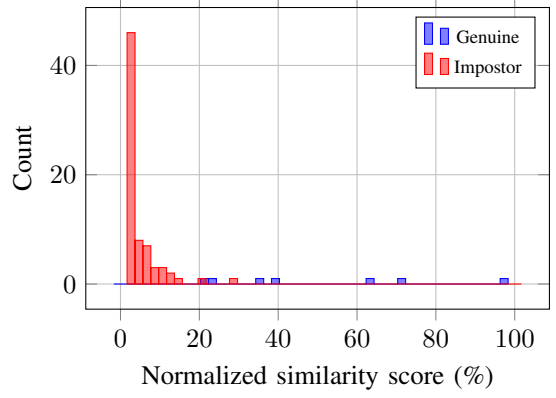


Figure 14: Score histogram for genuine and impostor comparisons.

templates while imposter considered as all the other combinations. For scoring methodology, we define the final score as the maximum similarity across all 5 user templates. For example, if the matching scores between a sample and the 5 probes are $[0.1, 0.3, 0.6, 0.9, 0.4]$, then the resulting score is 0.9.

We maintain same approach as Jain et al. [5], Cappelli et al. [7] and Slobodan et al. [15] and build histograms of the genuine and impostor combinations as shown in Figure 14. Then, we turn the histogram into a smoothed probability distribution function. We find the *FAR* and *FRR* and calculate *EER* as shown in Figure 15.

The most important outcome from this approach was to achieve similar *EER* of around 2% as achieved at industry and academical level sensors. [22], to objectively make a comparison with state-of-the-art algorithms. We calculate the mechanisms success probability as given in Equation 4 of a mechanism comprises a biometric credential and show that the optimal operating point is different from the *EER* point and achieves higher success probability as shown in Figure 16. Finally, we illustrate the probability success function of a mechanism composed of an atomic credential with $P_{\text{safe}}^1 = 0.9$ and $P_{\text{leak}}^1 = 0.04$, $P_{\text{loss}}^1 = 0.04$ and $P_{\text{stolen}}^1 = 0.02$ and integrate it with a biometric credential.

We show that the optimal operating point of the integrated mechanism is different from the *EER* point and achieves higher success probability by around 3% compared to the *EER* point, as the *EER* itself is only 1.8%. In addition, we can observe that the optimal point in both mechanisms achieves higher success probability than both the atomic credential mechanisms and biometric credential mechanism when tested individually as shown in Figure 16. Although there is an improvement in the success probability of the integrated mechanism, we can observe the improvement is not as significant as the one observed in the theoretical analysis. This can be explained by the fact that the *EER* point of the biometric credential is already very low, indicating high separability between a user and an attacker and thus, there is no much room for improvement. In that scenario, we compared several templates of user 103 against

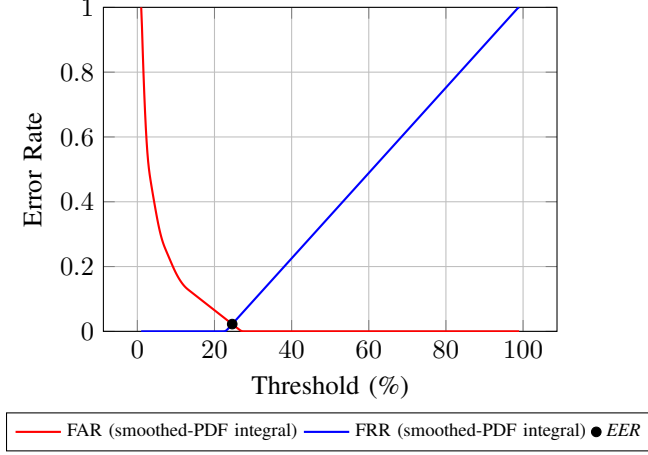
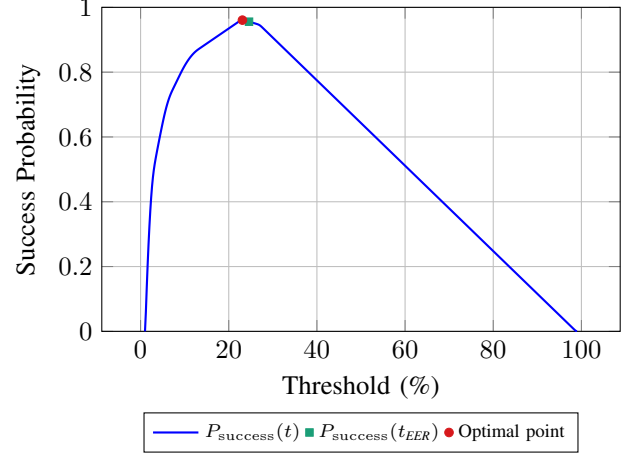


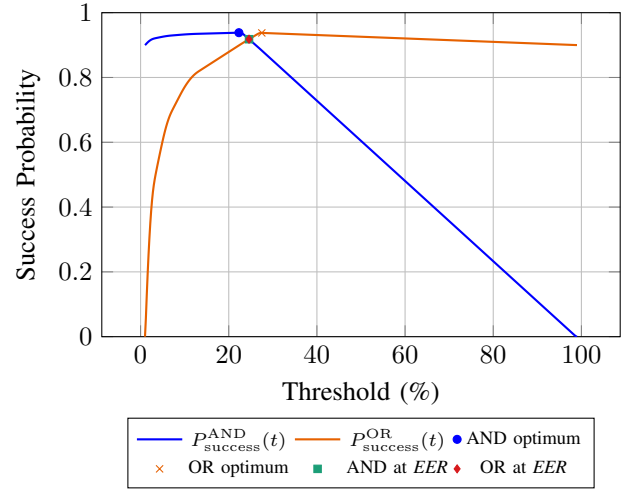
Figure 15: *FAR* and *FRR* as a function of the decision threshold, with the Equal Error Rate (*EER*) point.

templates of different users. However, our main approach in this paper was to increase the mechanism's success in scenarios where an imposter will try to spoof the user templates and authenticate into the user account. Hence, we expect the imposter to use spoofing tools, producing samples resamble with score distribution function that much more similar to the genuine distribution function, which might lead to increased *EER* point.

To support this claim, we use the LivDet2015 database [18], [19], [20], [21], which contains a large-scale live and spoof fingerprint samples. We follow the same methodology as before and used the SourceAFIS algorithm [17]. As probe templates, we select six Digital Persona scanner images of user 002_0. Each probe was matched against 14 additional images of the same user to generate genuine scores. For each target image, we define the final score methodology as the maximum similarity across 6 probe templates, similarly to the previous experiment. For example, if the matching scores between a target image and the six probes are [0.3, 0.5, 0.4, 0.8, 0.9, 0.7], then the resulting score is 0.9. To simulate a spoofing attack, we use fake fingerprint images of the same user (002_0) produced by *Gelatine*, *Woodglue*, *Latex* and *Ecoflex* and match them against the probe templates to obtain the imposter score distribution. We present the resulting histogram in Figure 17. We then compute the *EER* point and plot the *FAR-FRR* curve, as shown in Figure 18. As expected, the *EER* increased significantly to 17.8%. Finally, we calculate the success probability for both a single biometric credential and for a mechanism comprises an atomic credential and a biometric credential, as shown in Figure 19. We can observe that the optimal operating point reducing the failure probability of an *AND* mechanism by 60% and *OR* mechanism by 44% compared to the *EER* point. Moreover, we reduced the failure probability by 50% when integrating into *AND* mechanism compared to an atomic credential mechanism with $P_{\text{safe}} = 0.8$, $P_{\text{loss}} = 0.04$, $P_{\text{leak}} = 0.15$, $P_{\text{theft}} = 0.01$ probabilities. We observe that fixing the match threshold t to the *EER* when we integrate



(a) Single biometric credential mechanism's success.



(b) biometric credential integrated with an atomic credential into *AND* and *OR* mechanisms.

Figure 16: Success probability as a function of the threshold. The *AND* and *OR* mechanisms achieve success probabilities of 0.93 and 0.94 at the *EER* point, and 0.96 and 0.95 at their respective optimal operating points.

the biometric credential into the mechanism, significantly reduces its success probability, compared to the success probability of an atomic credential mechanism only.

8. Conclusion

We formalize the authentication mechanism model comprising, for the first time, both biometric and atomic credentials. We analyze the success of different mechanisms' types and present a new match threshold by modeling each credential with four states probabilities. We show we can improve any mechanism success probability when it comprises an atomic credential only, by integrating it with an asymptotically separable biometric credential. We conduct experiments using real public biometric datasets, simulating different scenarios include spoofing attack, which became

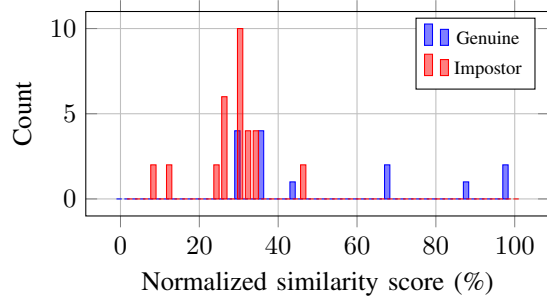


Figure 17: LivDet score histogram using the best score over probe templates.

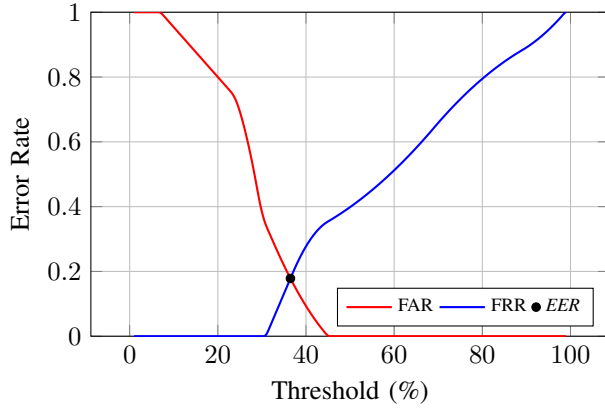


Figure 18: FAR and FRR as a function of the threshold using the best score over probe templates. The Equal Error Rate (EER) occurs at $\approx 17.8\%$.

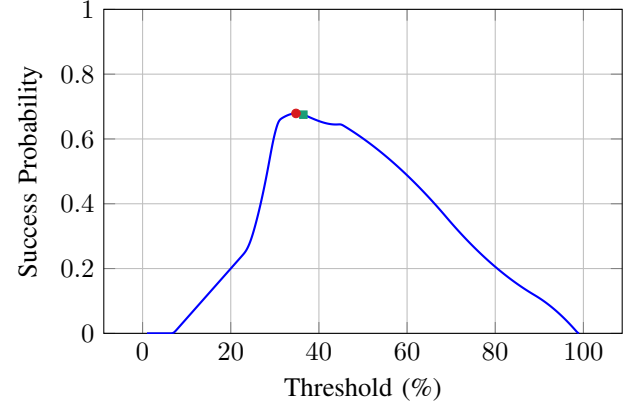
prevalant and easier on biometric credentials today. Our results show significant reduction of the failure probability when spoofing occurs.

Building new mechanisms protocols, based on the suggested match threshold (§4) is a promising direction for future work.

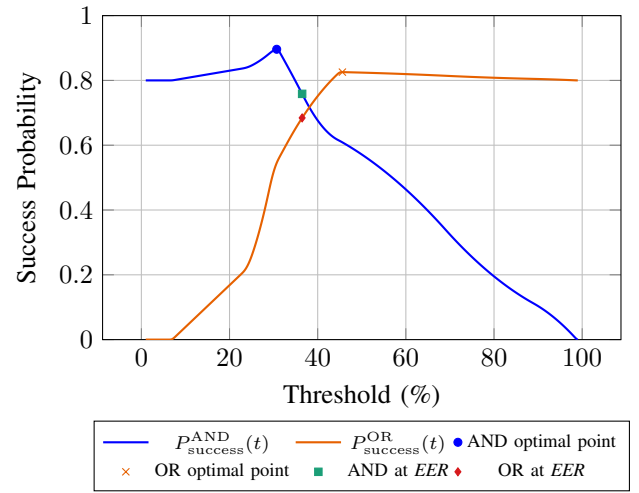
Our findings have important and impactful implications for the design of secure authentication mechanisms. We suggest letting the integrators to set the thresholds of the biometric credentials based on the other credentials they integrate into the mechanism. Harnessing our approach may affect millions of users around the world to enhance their security of digital and physical assets.

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(a) Single biometric credential.



(b) AND and OR mechanisms.

Figure 19: Success probability as a function of the decision threshold using the best score over probe templates. The AND mechanism achieves a maximal success probability of 0.896, while the OR mechanism achieves 0.826. At the EER operating point, the success probabilities are 0.759 and 0.684 for the AND and OR mechanisms, respectively.

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Appendix